M4. Reference Copies and a Shared ID Card

Week 3 OOP | Category B | Java solution with sample output

Objective

Implement the task using the classes, constructors, access control, references, or static members specified in the uploaded problem sheet.

Java Code

```
class IdCard {
    String name;
    int booksIssued;

    IdCard(String name, int booksIssued) {
        this.name = name;
        this.booksIssued = booksIssued;
    }
}

public class Main {
    public static void main(String[] args) {
        IdCard ravi = new IdCard("Ravi", 0);
        IdCard duplicate = ravi;    // Alias: does not create a new object.
        duplicate.booksIssued = 3;

        IdCard separate = new IdCard("Ravi", 3);

        System.out.println("Ravi's booksIssued (via first variable): "
                + ravi.booksIssued);
        System.out.println("duplicate == ravi: " + (duplicate == ravi));
        System.out.println("separate == ravi: " + (separate == ravi));
    }
}
```

Sample Output

```
Ravi's booksIssued (via first variable): 3
duplicate == ravi: true
separate == ravi: false
```

Concepts Covered

Reference copying, ==, object identity versus content equality